43008: Reinforcement Learning

Assignment 3, Part-B:
Design, Development/Implementation Plan

Project Title and number (Refer to Dashboard): Project #8: Cadence – Adaptive Building HVAC Control System

Team Name: No Limits

Team Members:

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Dataset/Environment Details (Optional, if available/planned):

Our group plans to use Sinergym, a gymnasium-based interface built on top of the EnergyPlus simulator. The library is compatible with the stable baselines 3 implementation of reinforcement learning algorithms and accepts a building model and weather data. The EnergyPlus simulator also comes with a host of building models and weather data that can be used for simulation. The table below summarizes the tools we plan to use:

Component	Details
Simulation Environment	Sinergym, a gymnasium-based wrapper built on top of the EnergyPlus simulation engine
Building Model	Our current choice is 5Zone, a single-story building divided into 5 zones (1 indoor/core and 4 perimeter) with an HVAC system. Has two actuators (heating setpoint, cooling setpoint) that operate at different temperature ranges.
Weather data	The EnergyPlus simulator also comes with several weather datasets, which will be used for training.
Reinforcement Learning Library	Stable Baselines 3. A library which contains implementations of RL algorithms, including **DQN** and **PPO**.

Interface/Deployment platform and plan:

The control agent is designed to be autonomous, with manual overrides by a human operator when necessary. We plan to implement an operations dashboard showing the building data, and the current setpoint being enforced by the agent, with a toggle to override the agent with a manual setpoint.

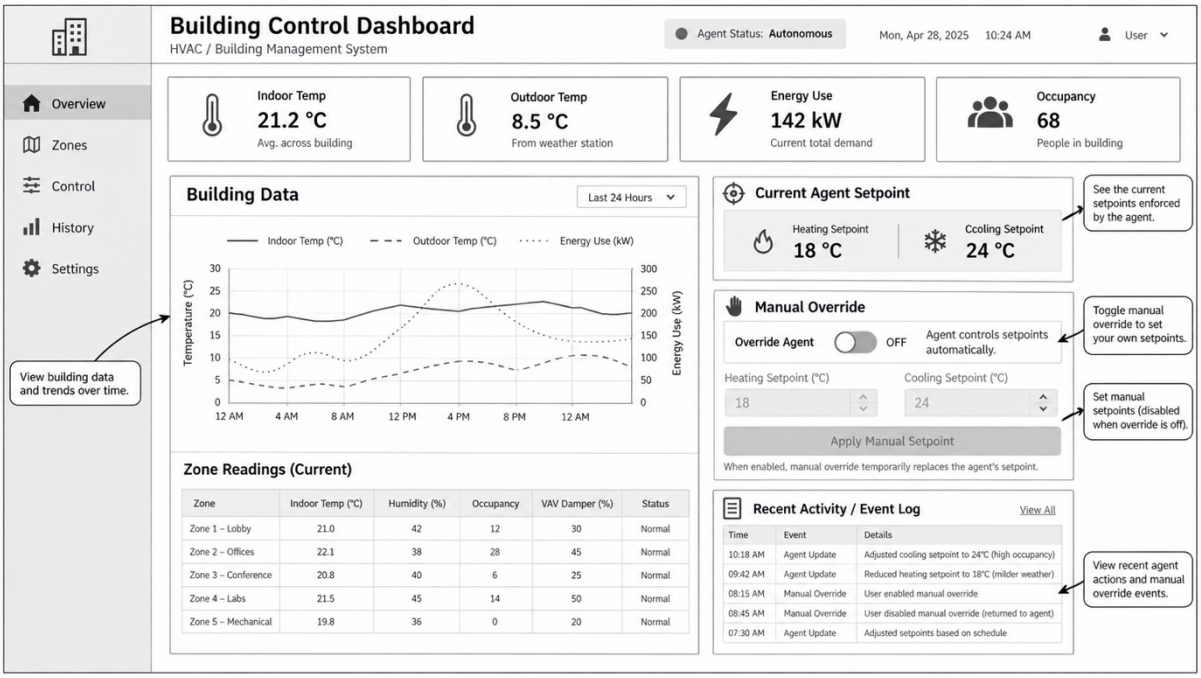


Figure 1: Building Operations Dashboard

Implementation plan:

The goal of this project is to create an agent that saves building energy consumption while maintaining comfortable conditions especially during operating hours. We primarily leverage Deep Q-Learning and Proximal policy optimization approaches and compare these with a fixed setpoint policy and a random policy. We hypothesize that the RL-based approaches will be able to develop more optimal building pre-conditioning policies based on weather conditions and operating hours than a simple rules-based approach. The following section further elaborates on the proposed implementation plan and estimated delivery timeline.

Training Strategy

The transition model of the environment is not known. Moreover, the observation space is continuous and high-dimensional, thus making model-based and tabular methods infeasible. In this problem, we explore two model-free approaches: Deep Q-Learning, and Proximal Policy Optimization.

The agent will be able to observe environment variables in 15-minute timesteps, each episode is a single year which is 35,040 timesteps. In each timestep, the agent will be able to observe the observation values which are shown in the table below. The discount factor will be configured to account for at least a day of operation.

Group	Variables
Time	month, day of month, hour
Outdoor	temperature, relative humidity, wind speed, wind direction, solar radiation rate (diffuse/direct)
Core Zone	temperature, relative humidity, occupant count
Setpoints	heating, cooling
Energy	CO2 emissions, HVAC electricity demand rate (W)
Meter	HVAC total electricity (J)

The HVAC system comes with two actuators: a heating setpoint, and a cooling setpoint. We plan to use a discretised action space, which has an explainability advantage to the end user as policy actions are clearly delineated. Sinergym provides a discretised controller configuration which is a combination of 10 heating and cooling setpoint combinations, discretised at roughly 1°C steps.

Action	Heating Setpoint	Cooling Setpoint
0	12°C	30°C
1	13°C	29°C
2	14°C	28°C
3	15°C	27°C
4	16°C	26°C
5	17°C	25°C
6	18°C	24°C
7	19°C	23.25°C
8	20°C	23.25°C
9	21°C	23.25°C

For reward calculation, we plan to use the default configuration provided by Sinergym and futrher introduce time-dependent thermal discomfort weighting as we want to penalize thermal discomfort more during the building's predefined operating hours. Sinergym's LinearReward class defines the reward signal as the following:

$$r_t = -\omega\, \lambda_P\, P_t - (1-\omega)\, \lambda_T\, (|T_t - T_{up}| + |T_t - T_{low}|)$$

- T_{up} and T_{low} define the upper and lower comfort range
- P_t is power consumption
- ω balances the energy consumption and thermal discomfort penalty trade-off
- λ is a scaling factor applied to the temperature and energy, so their magnitudes are relatively even

Metrics and Success Criterion

Metric	Description
Mean reward	Mean per timestep reward over the entire episode
Mean power demand	HVAC electricity demand rate (W)
Comfort violation (operational)	Percentage of time temperature is outside comfort range (building is within operational hours)
Unoccupied temperature excursion	Percentage of time temperate is outside comfort range (building is outside operational hours)

Effort Breakdown and Estimates

	Effort	Estimate
	Environment and evaluation configuration, fixed policy test and debugging	0.5 wk
Can do in parallel	DQN implementation, training, and debugging	1 wk
	PPO implementation, training, and debugging	1 wk
	GUI implementation	1 wk
	Model integration with GUI	0.5 wk
	Buffer for unexpected changes	1 wk
	Total	**5 weeks**